

Institute of Actuaries of India

Subject CT6 – Statistical Methods

March 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Zero sum game**

It is a mathematical model of a competitive situation between two parties, called players, who on each simultaneous move choose one of a number of possible actions. Each combination of actions result in a specified pay off and in zero sum game the sum of the pay offs for the two players is zero, i.e. what one player loses the other player wins.

Minimax solution is the solution which minimizes the maximum expected loss.

(2)**ii) if the man chooses Bus A first and Bus A goes to the correct destination then total travel time is 20 mins**

if the man chooses Bus A but it does not go to the destination then total travel time for to and fro journey is $20+20+15=55$ mins

similarly,

if the man chooses Bus B first and Bus B goes to the destination then total travel time is 15 mins

if the man chooses Bus B but it does not go to the destination then total travel time for to and fro journey is $15+15+20=50$ mins

The table below summarises the information:

Correct Bus	Man choice	
	A	B
A	20	50
B	55	15

It is two times more likely that bus A goes to the correct destination so if the man chooses A first then the expected travel time is

$$20 \times \frac{2}{3} + 55 \times \frac{1}{3} = 31.67 \text{ mins}$$

If the man chooses B first then the travel time is

$$50 \times \frac{2}{3} + 15 \times \frac{1}{3} = 38.33 \text{ mins}$$

In order to minimize the expected travel time the man should first choose bus A and the expected travel time is 31.67 mins.

(4)**[6 Marks]****Solution 2:**

$$i) Y_t = (1 + \lambda)Y_{t-1} - \lambda Y_{t-2} + e_t$$

$$\text{Or, } Y_t - Y_{t-1} = \lambda(Y_{t-1} - Y_{t-2}) + e_t$$

$$\text{Assume, } Z_t = Y_t - Y_{t-1}$$

$$\therefore Z_t - \lambda Z_{t-1} = e_t \approx N(0, \sigma^2)$$

Hence, $Z_t | Z_{t-1} \approx N(\lambda Z_{t-1}, \sigma^2)$

The likelihood function can be written as

$$L \propto \prod_{i=1}^n P(Z_i = z_i | z_{i-1}) P(z_0) = \prod_{i=1}^n \frac{e^{-\frac{(z_i - \lambda z_{i-1})^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma} \times 1$$

$$\text{Or, } L \propto \sigma^{-n} e^{-\frac{(z_i - \lambda z_i)^2}{2\sigma^2}}$$

$$\text{Taking log we get, } l = \log L = -n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (z_i - \lambda z_{i-1})^2 + C$$

Differentiating with respect to λ and equating to zero we get,

$$\frac{\partial l}{\partial \lambda} = \frac{1}{2\sigma^2} \sum_{i=1}^n 2z_{i-1}(z_i - \lambda z_{i-1}) = 0$$

$$\text{Or, } \hat{\lambda} = \frac{\sum_{i=1}^n z_{i-1} z_i}{\sum_{i=1}^n z_{i-1}^2}$$

Differentiating with respect to σ and equating to zero we get,

$$\frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (z_i - \lambda z_{i-1})^2 = 0$$

$$\text{Or, } \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (z_i - \hat{\lambda} z_{i-1})^2 \quad (5)$$

ii)

Using Yule-Walker equation we get,

$$\gamma_0 = \text{Cov}(Z_t, \lambda Z_{t-1} + e_t) = \lambda \text{Cov}(Z_t, Z_{t-1}) + \text{Cov}(Z_t, e_t) = \lambda \gamma_1 + \sigma^2$$

$$\gamma_1 = \text{Cov}(Z_{t-1}, \lambda Z_{t-1} + e_t) = \lambda \text{Cov}(Z_{t-1}, Z_{t-1}) + \text{Cov}(Z_{t-1}, e_t) = \lambda \gamma_0$$

From the above two equations we get,

$$\hat{\lambda} = \frac{\hat{\gamma}_1}{\hat{\gamma}_0} = \frac{\sum_{i=1}^n (z_i - \bar{z})(z_{i-1} - \bar{z})}{\sum_{i=1}^n (z_i - \bar{z})^2}$$

$$\text{And } \hat{\sigma}^2 = \hat{\gamma}_0 - \lambda \hat{\gamma}_1$$

To estimate λ in the second approach, we need to centralize the data around the mean of Z (which is first difference of Y). (5)

[10 Marks]

Solution 3:

i) Without any treaty the insurer pays the full amount of each loss
So the mean is

$$E(x) = e^{\mu + \frac{1}{2}\sigma^2}$$

$$= e^{9 + \frac{1}{2} \cdot 81} = 12,149$$

With treaty 1 the insurer pays the first 30,000 of every claim, so the net amount
 $X = \min(X, 30,000)$

$$E(x) = \int_0^{30,000} xf(x)dx + \int_{30,000}^{\infty} 30,000f(x)dx$$

Calculating the adjusted limits to apply the lognormal formulae to these integrals:

$$M_0 = (\log 30000 - 9) / 0.9 = 1.454$$

$$M_1 = M_0 - 0.9 = .554$$

Hence

$$E(x) = e^{\mu + \frac{1}{2}\sigma^2} (\phi(.554) - \phi(-\infty)) + 30,000(\phi(\infty) - \phi(M_0))$$

$$= 12,149(\phi(.554) + 30,000(1 - \phi(.554)))$$

$$= \text{Rs. } 10,817$$

(6)

ii) Expected amount paid by the reinsurer for treaty 1 = 12,149 - 10,817 = 1,331
Expected amount paid by the insurer for treaty 2

If the retained proportion is x% then the expected amount paid by the insurer is x% * 12,149

Hence

$$(1 - X\%) \cdot 12,149 = 1,331$$

$$X = 89\%$$

(4)

[10 Marks]**Solution 4:**

i) a) Inverse transform method

For a distribution function $F(x)$, Let $F^{-1}(y)$ be the inverse function to $F(x)$, defined for all y between 0 and 1. If a random variable U is uniformly distributed over the interval $[0, 1]$, then the random variable

$x = F^{-1}(U)$ has the distribution function $F(x)$ since:

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$$

So to generate a random variate X from the given distribution, we can use the following two step algorithm:

1. Generate a random variate u from $U(0,1)$
2. Return $x = F^{-1}(u)$ (2)

b) Acceptance rejection method

Acceptance rejection method is based on the idea that if it is difficult to generate points at random from a density f then a reasonable substitute might be to generate points from a larger area and discard any points which are not acceptable.

Having found a suitable density function $h(x)$ such that $f(x)/h(x)$ is bounded for all x define:

$$C = \max_{(all\ x)} f(x)/h(x) \text{ and } g(x) = f(x)/Ch(x) \text{ for all } x$$

So that $0 < g(x) \leq 1$

C represents the number of values which must be generated on average to end up with a single acceptable value of x .

Use the following algorithm:

1. Generate a random variate u from $U(0,1)$
2. Generate a random variate y from the density function $h(x)$
3. If $u > g(y)$ then go to step 1 otherwise return $x=y$ (3)

ii) Random numbers

X can take values $0, 1, 2, 3, \dots, n$.

The distribution function is given by

$$\begin{aligned} F(0) &= \binom{n}{0} 0.4^0 (1-0.4)^{(n-0)} \\ F(1) &= F(0) + \binom{n}{1} 0.4^1 (1-0.4)^{(n-1)} \\ F(2) &= F(1) + \binom{n}{2} 0.4^2 (1-0.4)^{(n-2)} \\ F(3) &= F(2) + \binom{n}{3} 0.4^3 (1-0.4)^{(n-3)} \\ F(4) &= F(3) + \binom{n}{4} 0.4^4 (1-0.4)^{(n-4)} \\ &\vdots \\ F(5) &= 1 \end{aligned}$$

If you have a random number U then return

$$\begin{aligned} X &= 0 \text{ if } 0 \leq U \leq F(0) \\ X &= 1 \text{ if } F(0) < U \leq F(1) \\ X &= 2 \text{ if } F(1) < U \leq F(2) \\ X &= 3 \text{ if } F(2) < U \leq F(3) \\ X &= 4 \text{ if } F(3) < U \leq F(4) \\ &\vdots \\ X &= n \text{ if } F(n-1) < U \leq 1 \end{aligned} \quad (3)$$

iii) Examples:**Sensitivity Analysis**

It is important to use a common set of random numbers for different simulations, otherwise the effect of changing the parameters for which sensitivity analysis is performed may be swamped by random effects arising from using a different set of random numbers.

Numerical evaluation of derivatives

It is important to use the same set of random numbers for calculation of outputs of random variables, otherwise the results will be distorted by variation of random numbers.

Comparative Simulation, performance evaluation

It is important to ensure same set of random numbers in order to avoid the possibility that the apparently superior performance of one scheme is due to a fortunate combination of simulated input random numbers.

(3)**[11 Marks]****Solution 5:**

- i) Since 'development year 3' is the ultimate year hence there will be no claims beyond 3 years

So, $C = 1600$

From accident year 2, we can calculate the development ratio for development year 2 to development year 3 as $2100/1800 = 1.16667$

Hence, B can be calculated as $= C / 1.16667 = 1600/1.16667 = 1371.43$

Now, from accident year 4, we can calculate the development ratio from development year 0 to development year 3 (ultimate) as $2800/1000 = 2.8$

So, the development ratio from development year 0 to development year 2 $= 2.8/1.16667 = 2.4$

The development ratios for first two development years are equal.

Hence, development ratio from development year 0 to development year 1 = development ratio from development year 1 to development year 2 $= 2.4^{0.5} = 1.549193$

Hence, E can be calculated as $= 1450 * 1.549193 * 1.16667 = 2620.72$

Now, development ratio from development year 0 to development year 1 $= (900 + D + 1450) / (A + 700 + 850) = 1.549193$ Eqn(1)

And development ratio from development year 1 to development year 2 $= (1371.43 + 1800) / (900 + D) = 1.549193$...Eqn(2)

From eqn (2), we get $D = 1147.15$

From eqn (1), we get $A = 707.40$

The final table is given by,

Accident Year	Development Year				Ultimate
	0	1	2	3	
1	707.40	900.00	1,371.43	1,600.00	1,600.00
2	700.00	1,147.15	1,800.00		2,100.00
3	850.00	1,450.00			2,620.72
4	1,000.00				2,800.00

(5)

ii) Earned premium for accident year 3 is 3000 and expected loss ratio is 80%.

Hence, E can be re-estimated as given below.

$$E = 1450 + 3000 \cdot 80\% \cdot (1 - 1/(1.549193 \cdot 1.16667)) = 2522.12$$

(1)

iii)

The complete cumulative claims amounts are given in the table below,

Accident Year	Development Year				Ultimate
	0	1	2	3	
1	707.40	900.00	1,371.43	1,600.00	1,600.00
2	700.00	1,147.15	1,800.00	2,100.00	2,100.00
3	850.00	1,450.00	2,179.26	2,522.12	2,522.12
4	1,000.00	1,549.19	2,400.00	2,800.00	2,800.00

Amount for 'Accident year 4 and development year 1' is calculated as = $1000 \cdot 1.549193 = 1549.19$

Amount for 'Accident year 3 and development year 2' is calculated as

$$= 2522.12 - 3000 \cdot 80\% \cdot (1 - 1/1.16667) = 2179.26$$

The non-cumulative outstanding claim amounts and corresponding reserves are given by,

Accident Year	Development Year				Outstanding reserve	Formula
	0	1	2	3		
1						
2				300.00	283.02	$= 300/1.06$
3			729.26	342.86	993.13	
4		549.19	850.81	400.00	1,611.17	$= 549.19/1.06 + 850.81/(1.06^2) + 400/(1.06^3)$
Total					2,887.31	

Hence the outstanding claim reserve is 2887.31.

(5)

[11 Marks]

Solution 6:

- i) Let X_{ij} denote the claim amount for insurance type i and year j .

From the data, we can calculate the below required functions.

Insurance Type	$\bar{X}_i = \left(\sum_{j=1}^3 X_{ij} \right) / n$	$\sum_{j=1}^3 (X_{ij} - \bar{X}_i)^2$	$(X_i - \bar{X})^2$
Two Wheeler	483.33	11,666.67	6,049.38
Car	383.33	11,666.67	493.83
Truck	350.00	5,000.00	3,086.42
Total	1,216.67	28,333.33	9,629.63

Now,

$$E[m(\theta)] = \bar{X} = 1216.67 / 3 = 405.56$$

$$E[s^2(\theta)] = N^{-1} \sum_{i=1}^N (n-1)^{-1} \sum_{j=1}^n (X_{ij} - \bar{X}_i)^2 = 28333.33 / 6 = 4722.22$$

$$V[m(\theta)] = (N-1)^{-1} \sum_{i=1}^N (X_i - \bar{X})^2 - n^{-1} E[s^2(\theta)] = 3240.74$$

$$\text{Hence, } Z = \frac{n}{n + \frac{E[s^2(\theta)]}{V[m(\theta)]}} = 0.6731$$

Expected claim outgo under Car Insurance for next year = $(1-Z)E[m(\theta)] + Z\bar{X}_2 = 390.60$ crores

Hence, total expected claim outgo under Car Insurance for next two years = $2 * 390.60 = 781.20$ crore.

(5)

ii)

The number of policies sold is assumed to be growing each year by 5%. Adjusted data needs to be considered to take account the above. Probable approaches can be as below:

1. In early years, we may consider lower volume of data by applying a Factor to each year and using EBCT model 2. If the Factor for year 4 is 1, then the corresponding factor for year 3 is $1/1.05$ and year 2 is $1/1.1025$ etc.
2. Increase the volume by 5% to be consistent with 'Year 4' values and then use the above method EBCT model 1.

(2)

iii)

Let Y_{ij} denote the claim amount for insurance type i and year j and P_{ij} denote number of policies sold for insurance type i and year j and X_{ij} is defined as Y_{ij}/P_{ij}

From the data, we can calculate the below required functions.

Insurance	$\sum_{j=1}^3 P_{ij} X_{ij}$	$\bar{P}_i = \sum_{j=1}^3 P_{ij}$	$\sum_{j=1}^3 P_{ij} (X_{ij} - \bar{X}_i)^2$	$\sum_{j=1}^3 P_{ij} (X_{ij} - \bar{X})^2$	$\bar{P}_i \left(1 - \frac{\bar{P}_i}{\sum_{i=1}^3 P_i} \right)$
Two Wheeler	1,450.00	1,900.00	12.25	18.11	1,115.22
Car	1,150.00	1,600.00	8.29	24.43	1,043.48
Truck	1,050.00	1,100.00	24.10	44.00	836.96
Total	3,650.00	4,600.00	44.65	86.54	2,995.65

Now, from definitions we get,

$$E[m(\theta)] = \sum_{i=1}^N \sum_{j=1}^3 P_{ij} X_{ij} / \sum_{i=1}^N \sum_{j=1}^3 P_{ij} = 3650 / 4600 = 0.7935$$

..(0.5)

$$E[s^2(\theta)] = N^{-1} \sum_{i=1}^N (n-1)^{-1} \sum_{j=1}^n P_{ij} (X_{ij} - \bar{X}_i)^2 = 44.65 / 6 = 7.4409$$

$$P^* = (Nn-1)^{-1} \sum_{i=1}^N \bar{P}_i \left(1 - \frac{\bar{P}_i}{\sum_{i=1}^3 P_i} \right) = 2995.65 / 8 = 374.4565$$

$$V[m(\theta)] = (P^*)^{-1} \left[(Nn-1)^{-1} \sum_{i=1}^N \sum_{j=1}^n P_{ij} (X_{ij} - \bar{X})^2 - E(s^2(\theta)) \right] = 0.009019$$

Now, credibility factor for Truck insurance is given by,

$$Z_3 = \frac{\sum_{j=1}^3 P_{ij}}{\sum_{j=1}^3 P_{ij} + \frac{E[s^2(\theta)]}{V[m(\theta)]}} = 0.5714$$

..(0.5)

Hence expected claim (in crores) per policy under Truck Insurance is given by = $(1 - Z_3)E[m(\theta)] + Z_3 \bar{X}_3 = 0.8855$

Expected claim in Year 4 under Truck Insurance = $500 * 0.8855 = 442.76$ crore

Expected claim in Year 5 under Truck Insurance = $600 * 0.8855 = 531.31$ crore

(8)
[15 Marks]

Solution 7:

i)

The log-likelihood function can be written as,

$$L = \text{Log}\left(\prod_{i=1}^n f(x_i)\right) = \log\left(\prod_{i=1}^n \lambda_i e^{-\lambda_i x_i}\right) = \sum_{i=1}^n (\log \lambda_i - \lambda_i x_i)$$

$$\text{Now, } \lambda_i = \begin{cases} e^a, & i \leq m \\ e^b, & \text{others} \end{cases} \text{ or } \log \lambda_i = \begin{cases} a, & i \leq m \\ b, & \text{others} \end{cases}$$

$$\text{Hence, } L = \sum_{i=1}^m (a - x_i e^a) + \sum_{i=m+1}^n (b - x_i e^b) = [-ma - (n-m)b + e^a \sum_{i=1}^m x_i + e^b \sum_{i=m+1}^n x_i]$$

Now, taking partial derivative w.r.t. a and b we get,

$$\frac{\partial L}{\partial a} = m - e^a \sum_{i=1}^m x_i$$

$$\frac{\partial L}{\partial a} = 0 \Rightarrow m - e^a \sum_{i=1}^m x_i = 0$$

$$\text{Hence, } \hat{a} = \log \left(\frac{m}{\sum_{i=1}^m x_i} \right)$$

$$\text{Similarly, } \frac{\partial L}{\partial b} = (n-m) - e^b \sum_{i=m+1}^n x_i \text{ gives}$$

$$\hat{b} = \log \left(\frac{n-m}{\sum_{i=m+1}^n x_i} \right)$$

Now scaled deviance is given by $2(L_f - L)$,

$$\text{Where, } L_f = - \sum_{i=1}^n \left(\log \left(\frac{1}{x_i} \right) - \frac{x_i}{x_i} \right) = - \sum_{i=1}^n \log x_i - n$$

Hence the scaled deviance is given by,

$$= 2 \left[- \sum_{i=1}^n \log x_i - n + \sum_{i=1}^m \left(\log \left(\frac{\sum_{i=1}^m x_i}{m} \right) + \frac{x_i}{\sum_{i=1}^m x_i} m \right) + \sum_{i=m+1}^n \left(\log \left(\frac{\sum_{i=m+1}^n x_i}{n-m} \right) + \frac{x_i}{\sum_{i=m+1}^n x_i} (n-m) \right) \right]$$

$$\begin{aligned}
&= 2 \left[- \sum_{i=1}^n \log x_i - n + \sum_{i=1}^m \left(\log \left(\frac{\sum_{i=1}^m x_i}{m} \right) \right) + m + \sum_{i=m+1}^n \left(\log \left(\frac{\sum_{i=m+1}^n x_i}{n-m} \right) \right) + n - m \right] \\
&= 2 \left[\sum_{i=1}^m \log \left(\frac{\frac{1}{m} \sum_{i=1}^m x_i}{x_i} \right) + \sum_{i=m+1}^n \log \left(\frac{\frac{1}{n-m} \sum_{i=m+1}^n x_i}{x_i} \right) \right] \quad (11)
\end{aligned}$$

ii)

The deviance residual is $\text{sign}(x_i - \hat{x}_i) \sqrt{D_i}$ where the deviance is $\sum_{i=1}^n D_i$

$$\hat{x}_i = \frac{150}{10} = 15$$

$$\begin{aligned}
D_1 &= 2 \left[- \log x_1 - 1 + \left(\log \left(\frac{\sum_{i=1}^m x_i}{m} \right) + \frac{x_1}{\sum_{i=1}^m x_i} m \right) \right] \\
&= 2 \left[- \log 10 - 1 + \left(\log \left(\frac{150}{10} \right) + \frac{10}{150} 10 \right) \right]
\end{aligned}$$

$$= 0.144264$$

$$\text{Hence the deviance residual is } = -\sqrt{0.144264} = -0.37982 \quad (3)$$

[14 Marks]

Solution 8:

i) a)

Following conditions must be satisfied by claims number process $\{N(t)\}_{t \geq 0}$

- 1) $N(0) = 0$
- 2) $N(s) \leq N(t)$ for $s < t$
- 3) $P(N(t+h)=r \mid N(t)=r) = 1 - \lambda h + o(h)$
- 4) $P(N(t+h)=r+1 \mid N(t)=r) = \lambda h + o(h)$
- 5) $P(N(t+h) > r+1 \mid N(t)=r) = o(h)$
- 6) For $s < t$, the number of claims in the time interval (s, t) is independent of the number of claims up to time s .

(3)

- b) Expected number of claims in two years will be 50, Mean and variance of the aggregate claims in two year period

$$E(S(2)) = 50 * e^{3 + \frac{1}{2} * 1.04} \\ = 1689.22$$

$$V(S(2)) = 50 * e^{6 + 2 * 1.04} \\ = 161461$$

Ruin will occur if $S(2)$ is greater than the premiums received and initial surplus

$$P[S(2) > 1000 + 1200 * 2] = P(N(0,1) > (3400 - 1689.22) / \sqrt{161461}) \\ = P(N(0,1) > 4.26) \\ = 1 - \Phi(4.26) \\ = .001\%$$

(3)

c) We need to calculate the premium for this we need

$$E(S(1)) = 844 \\ V(S(1)) = 80730 \\ \text{We need premium } P \text{ such that} \\ = P(1000 + P - S(1) < 0) = .05 \\ = P(S(1) > 1000 + P) = .05 \\ = P(N(0,1) < (P + 156) / 284) = .95 \\ = P + 156 = 1.644 * 284 \\ = 312$$

(3)

d) We need the loading $X\%$ such that
 $P[S(2) > (1 + x\%) * 1200 * 2 + 800 - 1000] = .05$
 $P(N(0,1) > (2200 + 2400x - 1689) / \sqrt{161461}) = .05$

$$2200 + 2400x - 1689 = 1.644 * 401 \\ X = 6\%$$

(3)

ii) Here, $p = 0.05$, $\lambda = 1/100 = 0.01$, Initial Capital = C

P (Ruin on First Claim)

$$= \sum_{t>0} P(\text{First Claim in year } t) P(\text{Claim} > C + 6 * t)$$

$$= \sum_{t>0} p q^{t-1} \times \int_{C+6t}^{\infty} \lambda e^{-\lambda x} dx$$

$$= \sum_{t>0} p q^{t-1} \times \left[-e^{-\lambda x} \right]_{C+6t}^{\infty}$$

$$= \sum_{t>0} p q^{t-1} \times e^{-6\lambda t} \times e^{-\lambda C}$$

$$= p \times e^{-6\lambda} \times e^{-\lambda C} \sum_{t>0} (q e^{-6\lambda})^{t-1}$$

$$= \frac{p \times e^{-\lambda(C+6)}}{1 - qe^{-6\lambda}}$$

$$\text{Now, } \frac{0.05 \times e^{-0.01(C+6)}}{1 - 0.95e^{-0.06}} < 0.05$$

Or, $C > 219.07$

Hence the minimum required capital requirement is 219.07 crore.

(6)

[18 Marks]

Solution 9:

i) GLM

There are three components of a generalised linear model;

A distribution of the data

A linear predictor which is a function of covariates

A link function which connects mean response to the linear predictor

(2)

ii) Let $Y \sim N(\mu, \sigma^2)$

Then;

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$$

$$= \exp\left\{\frac{-y^2 + 2\mu y - \mu^2}{2\sigma^2} - \log \sqrt{2\pi}\sigma\right\}$$

$$= \exp\left\{\frac{\mu y - 1/2\mu^2}{\sigma^2} - \frac{1}{2}\left(\frac{y^2}{\sigma^2} + \log 2\pi\sigma^2\right)\right\}$$

Comparing to $f(y) = \exp\left\{\frac{y\theta - b(\theta)}{a(\varphi)} + c(y, \varphi)\right\}$, which is the generalised form of exponential family of distribution, we get

$$\theta = \mu, b(\theta) = \frac{1}{2}\theta^2, \varphi = \sigma$$

$$a(\varphi) = \varphi^2 \text{ and } c(y, \varphi) = -\frac{1}{2}\left\{\frac{y^2}{\varphi^2} + \log 2\pi\varphi^2\right\}$$

Variance function and canonical link function

$$V(\mu) = b''(\theta) = 1$$

The canonical link function is $g(\mu) = \mu$

Therefore, the normal distribution belongs to an exponential family of distributions.

(3)

[5 Marks]
